GRACE COLLEGE OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING

Title

Calculating Family Expenses Using ServiceNow

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Abstract

This project titled “Calculating Family Expenses using ServiceNow” aims to develop a comprehensive, automated system for tracking and managing family expenses. The solution leverages the ServiceNow platform to design a dynamic, user-friendly interface where users can record daily expenses, categorize them, and generate summarized reports. The system enables smarter financial decision-making, encourages budget discipline, and enhances overall financial well-being.

Keywords

ServiceNow, Family Expenses, Expense Management, Business Rule, Financial Tracking, Budget Management

I. INTRODUCTION

Managing family expenses is often a manual and time-consuming process leading to errors and lack of visibility into financial patterns. The purpose of this project is to automate expense tracking through the ServiceNow platform — a robust, scalable, and cloud-based solution widely used for enterprise applications

This project introduces an Expense Calculation System that stores expense details, categorizes them, maintains relationships between daily and total expenses, and automatically updates reports. By integrating key ServiceNow features such as tables, forms, relationships, business rules, and automation scripts, the project provides an efficient solution for personal financial management.

II. OBJECTIVES

The primary objectives of this project, *“Calculating Family Expenses using ServiceNow,”* are as follows:

- **To design** an automated and reliable system for tracking and managing family expenses through the ServiceNow platform.
- **To create** well-structured relational tables that efficiently store and organize daily and overall expense data.
- **To implement** automated business rules that dynamically update total expenses and summaries in real-time.

- **To develop** a user-friendly and structured interface that enables easy data entry, tracking, and analysis.
- **To generate** accurate, real-time reports that help users monitor their financial habits and make informed budgeting decisions.

III. SYSTEM REQUIREMENTS

- Hardware Requirements
  - Processor: Intel i3 or above
  - RAM: Minimum 4 GB
  - Storage: 100 MB of free space
  - Internet Connection: Required
- Software Requirements
  - Platform: ServiceNow Personal Developer Instance
  - Browser: Google Chrome or Microsoft Edge
  - Programming/Scripting: JavaScript (for Business Rules)

IV. METHODOLOGY

The methodology adopted for the project *"Calculating Family Expenses using ServiceNow"* is based on a systematic, modular, and automation-oriented approach. Each stage was carefully executed to ensure accuracy, maintainability, and scalability of the system.

Step 1 – Setting up ServiceNow Instance

First, I signed up for a developer account on the ServiceNow Developer website

Then I requested a personal developer instance and received the login credentials by email.

After that, I logged into my ServiceNow instance and started working.

Step 2 – Creating a New Update Set

In this step, I searched for "Local Update Set" in the filter navigator, clicked on "New,"

and entered the name *Family Expenses*.

Then I submitted and made it current, so all the configurations are stored in this update set.

Step 3 – Creating Family Expenses Table

Next, I went to Tables and created a new table named *Family Expenses*.

This table is used to store the main expense details.

I also created a new menu called *Family Expenditure* to access it easily.

Step 4 – Creating Columns for Family Expenses Table

I added several fields:

- Number (String type)
- Date (Date type)
- Amount (Integer type)
- Expense Details (String type, with maximum length 800).

Then I saved the table.

Step 5 – Making Number Field Auto-Generated

I made the Number field auto-generated using the dynamic default value "Get Next Padded Number."

Then I created a Number Maintenance record with prefix *MFE* for Family Expenses.

Step 6 – Configuring the Form

I opened the Family Expenses form in form designer, rearranged the fields, made the Number field read-only, and set Date and Amount fields as mandatory.

Then I saved the form design.

Step 7 – Creating Daily Expenses Table

Next, I created another table called *Daily Expenses* under the same Family Expenditure menu.

This table will store each day's individual expense details.

Step 8 – Creating Columns for Daily Expenses Table

Here I added the following fields:

- Number (String)
- Date (Date)
- Expense (Integer)

- Family Member Name (Reference type)
- Comments (String, max length 800).

Then I saved the table.

Step 9 – Auto-Number for Daily Expenses

Again, I set the Number field as auto-generated using “Get Next Padded Number”

and created a Number Maintenance entry with prefix **MFE**.

Step 10 – Configuring the Form

I customized the Daily Expenses form layout,

made the Number field read-only,

and made Date and Family Member Name fields mandatory.

Then I saved the design.

Step 11 – Creating Relationship Between Tables

Now, I created a relationship between **Family Expenses** and **Daily Expenses** tables.

This helps to link each family expense with its related daily expenses.

Step 12 – Configuring Related List

I opened Family Expenses and configured the related list to show **Daily Expenses** entries linked by date.

This way, I can see all daily expenses for a particular date under one main record.

Step 13 – Creating Business Rule

I created a Business Rule on the Daily Expenses table.

This rule automatically updates the Family Expenses table whenever a new daily expense is added or updated.

It adds the amount to the total and updates the expense details text.

This helps in automatic calculation and summary.

Step 14 – Final Relationship Query

Finally, I added a script in the relationship query section so that the Daily Expenses records

are filtered and displayed only for the same date as the Family Expenses record.

V. RESULTS AND DISCUSSION

The project "Calculating Family Expenses using ServiceNow" efficientl automates expense tracking and reporting. It accurately computes total expense from daily records and ensures real-time synchronization between related tables The system provides clear visualization of spending patterns through an intuitiv interface, making financial management simple and reliable. Overall, th implementation demonstrates ServiceNow's flexibility beyond ITSM, proving it potential for effective personal and family expense management

VI. CONCLUSION

The project *"Calculating Family Expenses using ServiceNow"* proves tha ServiceNow can effectively manage daily financial activities beyond it traditional ITSM scope. The system automates expense tracking, total calculation and report generation through a simple, user-friendly interface. Overall, i highlights the platform's adaptability and potential for innovative persona finance applications.

VII. OUTPUT LINKS

Drive Video Link:
https://drive.google.com/file/d/1QBwREnmyJ05jNzaiGI6bfL7UKO3tAfuj/view?usp=drivesdk

GitHub Link:https://github.com/krishnan-ctrl/Calculating-Family-Expenses-using-ServiceNow/